

Particulate Matter Monitoring & Health Effects

Issue 2026-09-02 | Entry window 2 September 2026 (single day) | 15 records in scope, 62 screened out

15RECORDS IN SCOPE
(62 SCREENED OUT)**6**SUBTOPIC CLUSTERS
OF TEN OCCUPIED**4**EFFECT ESTIMATES
(TWO SOURCES)**6**RECORDS SHIPPING
ON METADATA ONLY**9**RECORDS WITH NO
HEALTH ENDPOINT

Signal of the day

A national STEMI register and a Mendelian-randomisation study land on the same day at opposite ends of the effect-size scale — and the gap between them is the whole methodological problem of PM epidemiology in one figure.

De Potter et al. analysed **every ST-elevation myocardial infarction admission in Belgium** over nine years ($n=51,544$) in a time-stratified case-crossover design with residential exposure from high-resolution models. Per IQR, same-day $PM_{2.5}$ gave **OR 1.022 (1.008–1.035)** — a 2% acute excess, tight CI, and the kind of number a design that self-matches on person and season is built to produce. Li H et al. instead instrument lifetime $PM_{2.5}$ genetically and report **OR 1.897 (1.063–3.388)** for childhood asthma, with **FUT10 mediating 19.3%** of the effect and 70 TWAS-implicated genes enriched in lysosomal and leukocyte-mediated immunity. Both are defensible; they are not measuring the same thing. The case-crossover isolates a trigger effect over hours with the chronic exposure held constant by design; the MR estimates a lifetime contrast with the acute component invisible. Reading either as “the” PM effect is a category error, and any sensor network that claims to serve epidemiology has to declare which of the two it is resolving — minutes-to-days for the first, cumulative annual burden for the second. They imply completely different siting, duty cycle and calibration requirements.

What changed today

- **Six of fifteen records ship on metadata only** — the worst abstract availability this watch has recorded. Elsevier deposited five *Atmos. Environ., Environ. Pollut., Sci. Total Environ.* and *Build. Environ.* records to Crossref with no abstract, none is yet in PubMed or Europe PMC, and Semantic Scholar returns 404 or a null abstract for all of them. They are counted and listed; **no number is attributed to any of them**, and four are on-topic enough ($PM_{2.5}$ -bound metals, solid-fuel PAH biomarkers, heritage-site particle sensing, landfill receptor monitoring) that they belong on the abstract retry list. All six are scored *Metadata only* in the architecture figure rather than being given a design guessed from the title.
- **Two effect estimates on the same acute contrast, in the same paper, disagree by pollutant.** De Potter et al. report same-day $PM_{2.5}$ **OR 1.022 (1.008–1.035)** against NO_2 **OR 1.044 (1.025–1.063)** per IQR, with O_3 strongest at lag 0–2 (**1.063, 1.034–1.090**). NO_2 doubling the $PM_{2.5}$ coefficient in a single-pollutant comparison is the traffic-proximity confounding this literature never resolves — and in-hospital mortality (11.8%) was driven not by any pollutant but by individual and area income and retirement status.
- **A filtration material with a number worth writing down.** Lv H et al.’s single-component polyimide gradient aerogel reaches **99.996% $PM_{0.3}$ efficiency at 45.08 Pa initial pressure drop with 127 g m^{-2} dust-holding capacity**, holding 99.91% at 300 °C. The gradient architecture buys depth filtration rather than surface caking, which is why the dust-holding number is the interesting one, not the efficiency.
- **Personal exposure during exercise, measured rather than modelled.** Engeroff et al. ran 60 instrumented 2 km runs across a park, a residential street and a major road at two times of day. Lowest PM across all fractions: **midday in the park**; highest PM_{10} : **morning on the same park route**. Route choice and rush-hour avoidance both work, but the meteorological covariates (humidity, pressure, wind, temperature) carried real signal — which is a warning about drawing route conclusions from a single campaign.
- **Sulfate aerosol water turns out to be photochemically active on its own.** Petersen et al. show the sulfate anion absorbs surface sunlight and ejects an electron to form $SO_4^{\bullet-}$ on a sub-microsecond timescale, at rates **comparable to measured hydroxyl-radical production in aerosols**. Biomass-fire chromophores photosensitise the pathway, raising radical formation **up to five-fold** and more strongly at low pH. The consequence is not marginal: secondary brown carbon from this chemistry **flips the direct radiative forcing of sulfate-containing aerosol from negative to positive**.
- **No sensing-side calibration or drift work today.** *Sensing, forecasting & instrumentation* holds two records, both metadata-only. OpenAlex was 403 and arXiv 429 again, Consensus was not re-queried after returning three already-seen records on the 1 September run, and PubMed’s

instrumentation axis returned **zero** hits for this entry date.

Corpus analytics

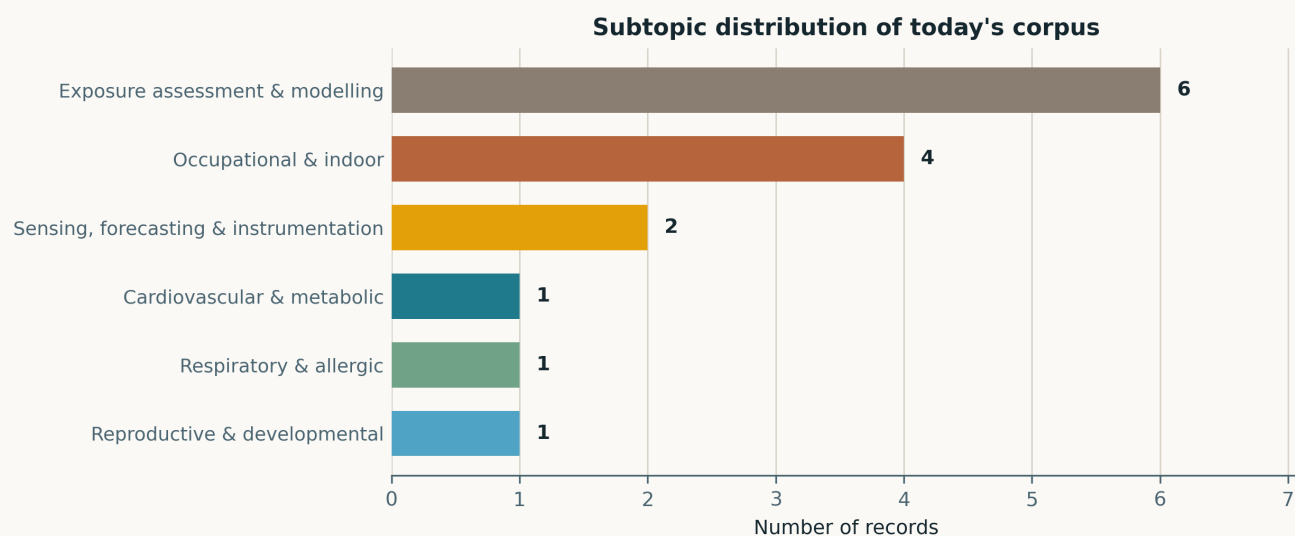


Fig. Subtopic assignment of the 15 in-scope records. **Only six of ten clusters are occupied** — *Mechanistic toxicology*, *Neuro / mental health*, *Burden, policy & mitigation* and *Other clinical endpoints* are all empty, the first day since 24 August with four or more simultaneously vacant. *Occupational & indoor* at 4 is unusually large for this watch and is genuinely heterogeneous: a filtration material, a surgical-smoke bench test, a solid-fuel biomarker study and an infiltration metric.

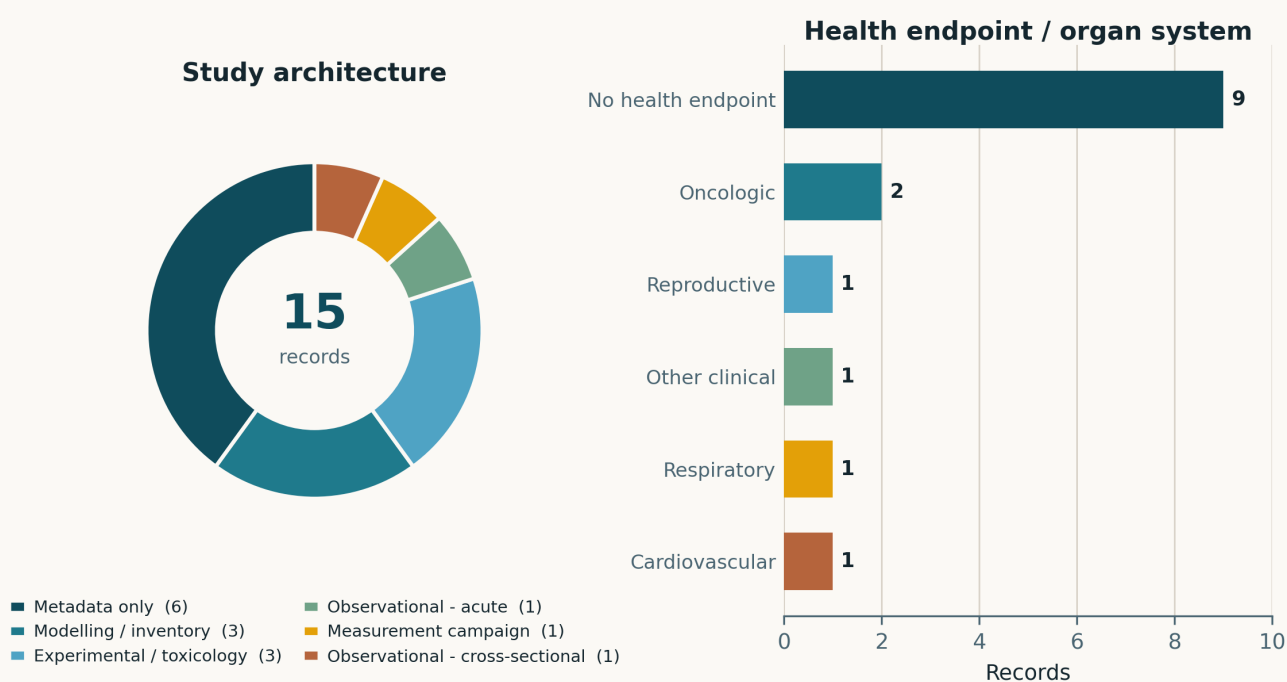


Fig. Left — study architecture. *Metadata only* is the largest group at 6 of 15 — these records are scored as an absence of information rather than assigned a design inferred from their titles, which is why the real architectures behind them do not appear anywhere in this panel. Of the nine that do carry a design, *Modelling / inventory* and *Experimental / toxicology* hold 3 each and the single *Observational — acute* entry is the Belgian case-crossover. Right — health endpoint. **Nine of fifteen (60%)** carry no human health endpoint, below 1 September's 67%; four of the six that do are metadata-only or MR-derived, so the observed-incidence content of this issue is one paper.

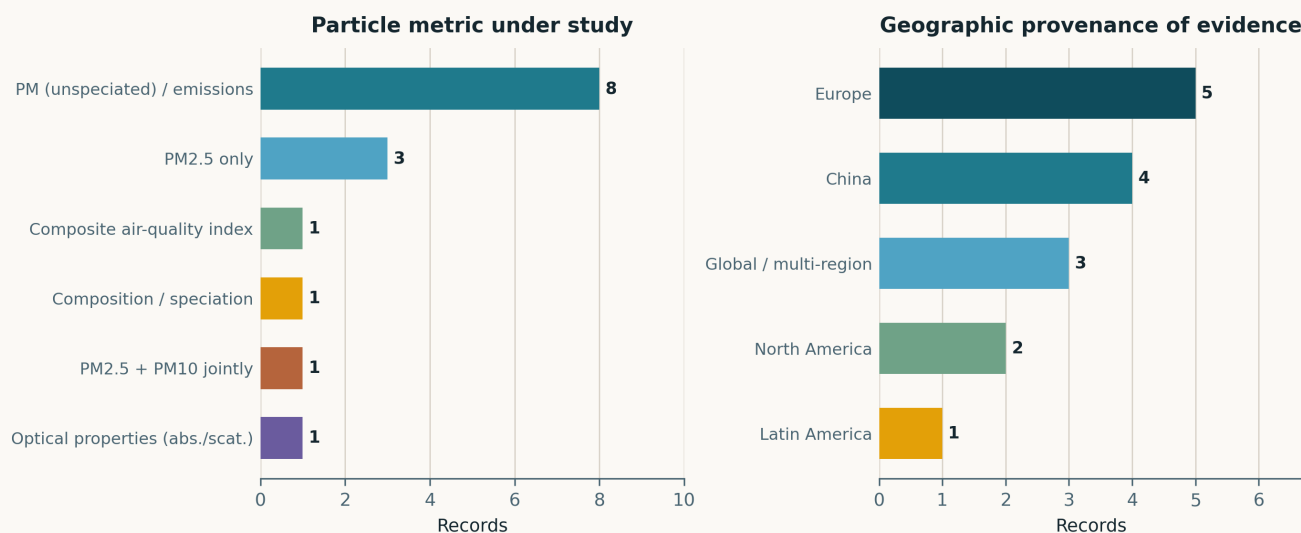


Fig. Left — particle metric. The eight records in *PM (unspecified) / emissions* are inflated by the metadata-only entries, whose particle metric can only be inferred from the title; treat that bar as an artefact of missing abstracts rather than a statement about the day's measurement practice. Right — geographic provenance. Europe leads at 5 — Belgium and Germany named, three coded only to *Europe* because two are multi-country and one ships on metadata — then China 4, North America 2, Latin America 1. **Sub-Saharan Africa, South Asia, Southeast Asia and Oceania are all absent**; Latin America returns for the first time since 28 August, via a heritage-conservation sensing study.

Life-course windows addressed today (records may span several stages)

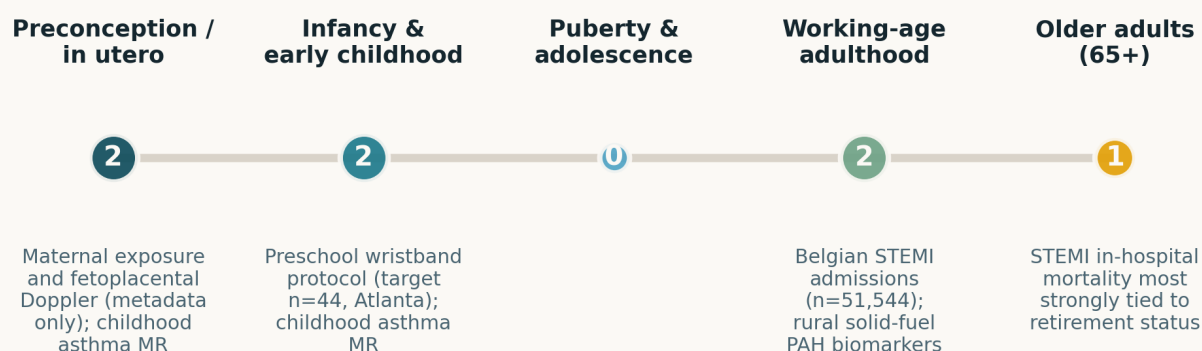


Fig. Life-course windows addressed; counts are non-exclusive. **Puberty and adolescence is empty.** The childhood-asthma MR is scored against both the in-utero and early-childhood cells because a genetically instrumented lifetime exposure has no identifiable window — which is precisely the interpretive weakness of the design, and scoring it in a single cell would hide that.

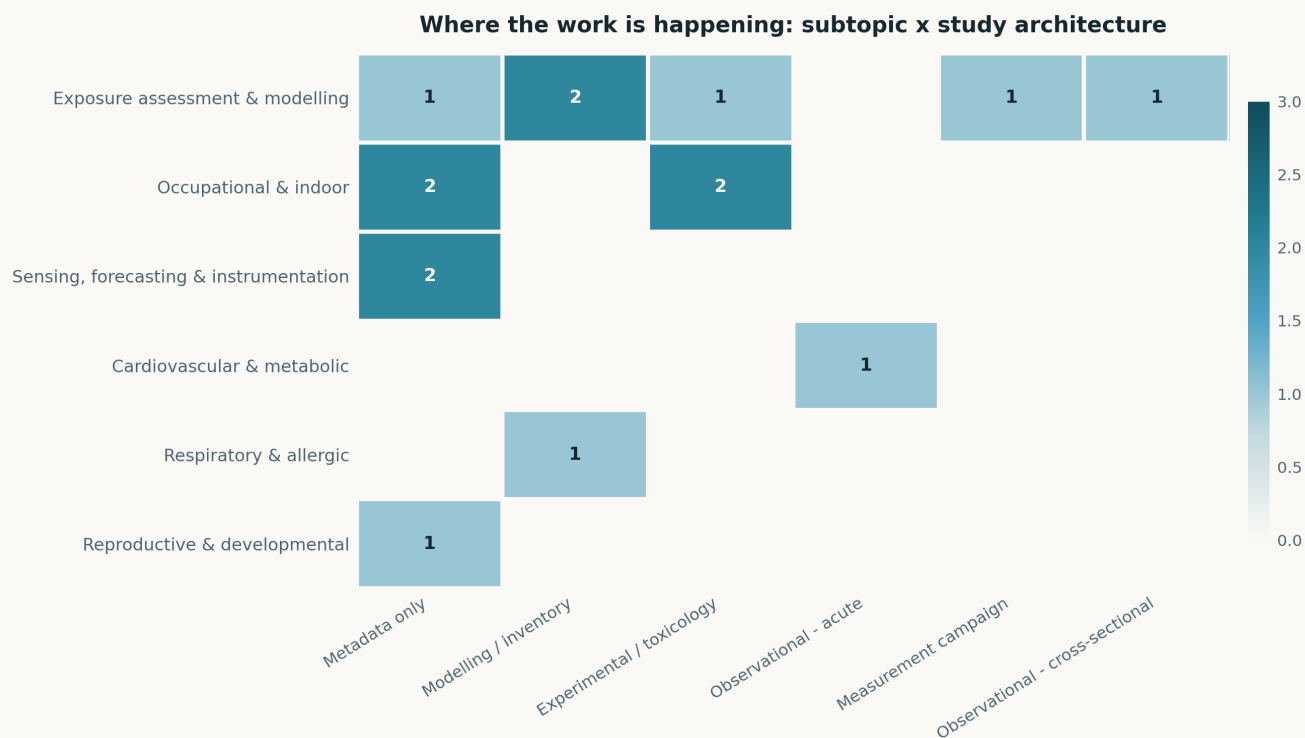


Fig. Subtopic × study-architecture cross-tabulation. Sparse by construction at $n=15$, and the *Metadata only* column carries 6 of the 15 — including *both Sensing, forecasting & instrumentation* records, which is why that row contributes no method detail at all. Read this figure as a map of what is missing as much as of what is present.

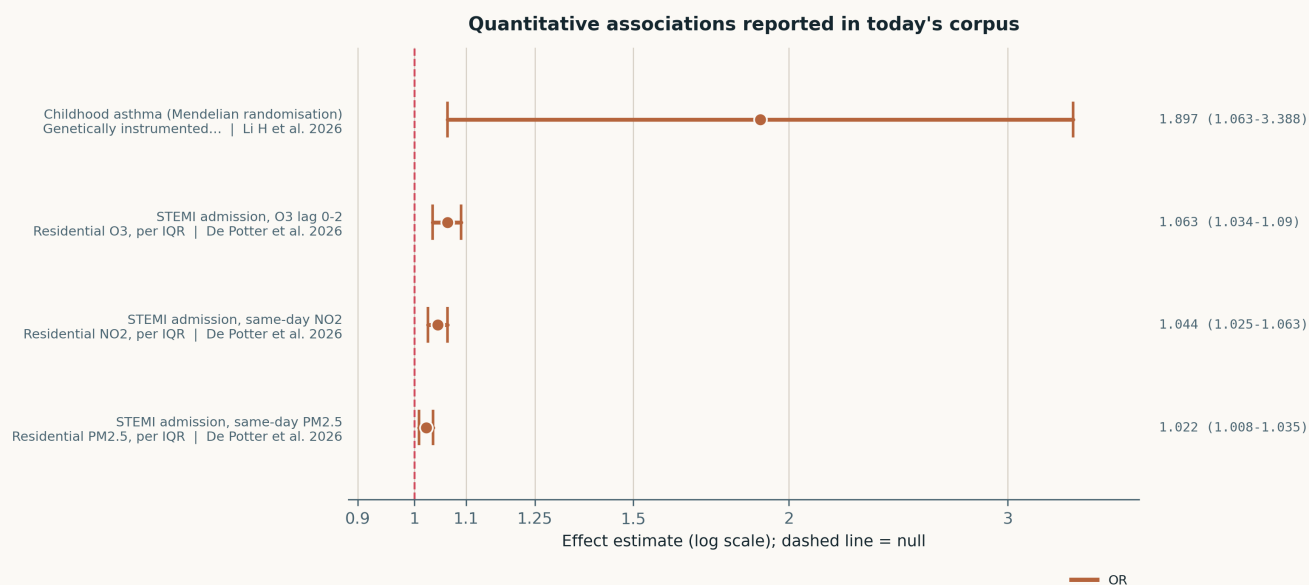


Fig. Four estimates from **two** sources. The three Belgian case-crossover odds ratios are per-IQR, same-day (or lag 0–2 for O₃) contrasts within one register and are not independent of one another; the childhood-asthma odds ratio is a Mendelian randomisation on a lifetime genetically instrumented exposure. They share an axis but not a scale of inference — the MR point estimate is roughly forty times larger in excess risk (89.7% vs 2.2%) and its interval spans a factor of three. Do not read the spread as heterogeneity between studies of the same quantity.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Cardiovascular & metabolic

1 record

De Potter et al. 2026 Eur Heart J Acute Cardiovasc Care

Socioeconomic and environmental determinants of STEMI incidence and in-hospital mortality

Nine years of Belgian national hospital discharge data, **51,544 STEMI admissions** (51.4 per 100,000 per year), with residential PM_{2.5}, NO₂, O₃, temperature, noise and greenspace from high-resolution models. A time-stratified case-crossover handled the acute question and multivariable logistic regression the mortality question. Per IQR, same-day: **PM_{2.5} OR 1.022 (1.008–1.035)**, **NO₂ OR 1.044 (1.025–1.063)**; O₃ peaked over lag 0–2 at **1.063 (1.034–1.090)**. In-hospital mortality was 11.8% and tracked *socioeconomic* variables — lower individual and area income, retirement status — rather than any pollutant. Two readings matter: NO₂ outperforming PM_{2.5} in single-pollutant models is the standard traffic-proximity confounding, and the incidence/mortality split says exposure and outcome severity are governed by different mechanisms.

doi: 10.1093/ehjacc/zuag118

Reproductive & developmental

1 record

Zanini et al. 2026 Environ Pollut

Maternal air-pollution exposure and fetoplacental Doppler markers

Ships on metadata only. The Crossref deposit carries no abstract, the record is not in PubMed or Europe PMC, and Semantic Scholar returns 404 for the DOI, so cohort size, exposure model and Doppler effect sizes are all unavailable. Counted and listed; no numbers claimed. Flagged for the abstract retry list — fetoplacental haemodynamics is a mechanistically informative intermediate between maternal PM exposure and birth outcome and is thinly represented in this watch's corpus.

doi: 10.1016/j.envpol.2026.129046

Respiratory & allergic

1 record

Li H et al. 2026 ERJ Open Res

Multi-omics Mendelian randomisation of PM_{2.5} and childhood asthma

Two-sample MR on large-scale GWAS summary data gives **OR 1.897 (1.063–3.388)**, $p=0.030$ for genetically instrumented PM_{2.5} on childhood asthma. A TWAS identified **70 genes** co-expressed with both, enriched in lysosome- and leukocyte-mediated immunity, with **MEAF6** validated as protective and **RNF40** as a risk gene. Two-step MR put **FUT10 as a positive mediator carrying 19.3%** of the effect, with CD200 and MANBA negative mediators; a phenome-wide MR flagged associations of these targets with other diseases, which the authors correctly present as a caution against treating them as drug targets. The limits are the usual MR limits and they are severe here: the PM_{2.5} instrument is a genetic proxy for a residentially correlated environmental exposure, so horizontal pleiotropy through socioeconomic position is not excluded, and the interval spans a factor of three.

doi: 10.1183/23120541.01444-2025

Occupational & indoor

4 records

Lv H et al. 2026 ACS Appl Mater Interfaces

Polyimide microsphere/nanofibre gradient aerogels for high-temperature filtration

A single-component polyimide gradient aerogel built to beat the dust-holding limit of uniform nanofibre mats, which cake at the surface and choke. Dual-layer structure: ≈ 730 nm coarse fibres under ≈ 300 nm medium fibres, with a composite functional layer of ≈ 3.34 μm microspheres and ≈ 60 nm ultrafine fibres on top. Performance: **99.996% PM_{0.3} efficiency at 45.08 Pa** initial pressure drop, **127 g m⁻² dust-holding capacity**, and **99.91% retained at 300 °C**. Single-component construction is claimed for recyclability and uniform thermal expansion. What is not reported is loaded pressure-drop evolution or efficiency after thermal cycling, which is what actually determines service life in a stack.

doi: 10.1021/acsami.6c10042

Ma L et al. 2026 Can J Urol

Surgical smoke from laser laparoscopic partial nephrectomy with an aspirating handle

Ex vivo porcine kidney comparison of electrocoagulation, ultrasonic, 980 nm diode and 532 nm KTP laser resection, with and without a novel aspirator-integrated handle. The KTP green laser produced the most smoke; coagulation-layer thickness ran diode (**4.16 ± 0.21 mm**) > green (1.55 ± 0.21) > electrocoagulation (0.35 ± 0.08) > ultrasonic (0.52 ± 0.08). With the aspirator, diode-laser formaldehyde fell **4.95 → 1.69 mg m⁻³**, TVOC **8.34 → 1.11 mg m⁻³** and CO **124 → 24 ppm**. Gases only — the study reports no particle-size distribution or number concentration, which is the fraction that reaches the alveoli and the reason surgical smoke is an occupational aerosol problem rather than a gas one. In vitro, no personal sampling.

doi: 10.32604/cju.2026.070096

Chen Y et al. 2026 Atmos Environ

Solid-fuel PAH exposure, phenanthrene metabolites, DNA damage and inflammation

Ships on metadata only — no abstract in the Crossref deposit and no PubMed, Europe PMC or Semantic Scholar record on this run. Title and authorship indicate urinary phenanthrene metabolites as the exposure index in a rural northern Chinese solid-fuel population, with DNA-damage and systemic-inflammation biomarkers as outcomes. Counted and listed; no numbers claimed. High priority for the retry list: household solid-fuel combustion is the single largest PM exposure contrast available in observational work and this watch carries very little of it.

doi: 10.1016/j.atmosenv.2026.122339

Micallef & Scerri 2026 Build Environ*A flow-conditioned infiltration metric for traffic-related pollution ingress*

Ships on metadata only — Crossref carries no abstract and the record is not otherwise indexed on this run. The title indicates an outdoor-to-indoor infiltration metric for traffic-related air pollution in naturally ventilated buildings, conditioned on flow regime rather than assumed constant. Counted and listed; no numbers claimed. Worth retrying: a flow-dependent infiltration factor is exactly the term that makes outdoor-monitor-based exposure assignment wrong for naturally ventilated stock, and it is the term a paired indoor/outdoor sensor pair can actually estimate.

doi: 10.1016/j.buildenv.2026.115195

Sensing, forecasting & instrumentation**2 records****Chaparro et al. 2026** Sci Total Environ*Biological and technological sensors for airborne particles at heritage sites*

Ships on metadata only. The title indicates parallel biomonitoring and instrumental sensing of airborne fine particles for preventive conservation of cultural heritage, but no abstract was deposited and the record is not in PubMed, Europe PMC or Semantic Scholar on this run, so the sensor models, the biomonitor species and any agreement statistics between the two are unavailable. Counted and listed; no numbers claimed. On the retry list — cross-validation of a biological against an instrumental particle sensor is directly relevant to low-cost network validation and rarely published.

doi: 10.1016/j.scitotenv.2026.182285

Delkash 2026 Environ Pollut*Barometric pressure variability in community air monitoring near a landfill*

Ships on metadata only — the Crossref deposit has no abstract and Semantic Scholar returns the record with a null abstract field. The title describes a receptor-scale multi-method analysis of how barometric pressure variability imprints on community monitoring data around an elevated-temperature landfill, across two mitigation interventions. Counted and listed; no numbers claimed. This is the single most operationally relevant record of the day for network work — pressure-driven emission modulation is a known confounder in fenceline and community monitoring and is almost never quantified — so it goes to the top of the retry list.

doi: 10.1016/j.envpol.2026.129084

Exposure assessment & modelling**6 records****Petersen et al. 2026** Environ Sci Technol*Sulfate ions in aerosol water are their own source of radical oxidants*

Direct spectroscopic evidence that the sulfate anion absorbs surface-level sunlight and ejects an electron, forming $\text{SO}_4^{\bullet-}$ **on a sub-microsecond timescale** in a concentration-dependent manner; long-timescale production rates quantified by radical trapping with LC-HRMS. Compounds emitted from biomass fires photosensitise the pathway, raising radical formation **up to five-fold** over direct photolysis across pH 1.6–5.5, more strongly at low pH. Estimated $\text{SO}_4^{\bullet-}$ production is **comparable to measured hydroxyl-radical production in aerosols** — i.e. a second oxidant budget nobody was counting. The downstream claim is the striking one: secondary brown carbon from this chemistry **turns the direct radiative forcing of sulfate-containing aerosol from negative to positive**, and the organosulfates formed are cloud-active. Bulk solutions at controlled ionic strength; extrapolation to real aerosol water is the open step.

doi: 10.1021/acs.est.6c05704

Engeroff et al. 2026 Research Square (preprint)*PM exposure while running: park, residential and roadside routes by time of day*

Mobile personal PM_{10} , $\text{PM}_{2.5}$ and PM_1 over **60 instrumented 2 km runs** (20 each on a major road, a residential street and an urban park; 30 morning, 30 midday; pace 8:02–4:31 min/km) with local urban background as reference and linear mixed-effects models adjusting for meteorology. Lowest concentrations of every fraction: **midday in the park**. Highest PM_{10} anywhere: **morning on that same park route**. Exposure across all routes was lower at midday, tracking urban background, and humidity, pressure, wind speed and temperature all carried signal. The conclusion — park routes and rush-hour avoidance are sufficient — is reasonable but rests on 20 runs per route in one city, and the fact that the park produced both the minimum and the maximum should temper any general route advice. Preprint, not peer reviewed.

doi: 10.21203/rs.3.rs-10016764/v1

Zhu et al. 2026 Atmos Chem Phys*Stabilized Criegee intermediates and the sensitivity of sulfuric acid formation*

Criegee reactions in MCM v3.3.1 updated to current IUPAC recommendations, driven through a large AtChem box-model ensemble, with interpretable XGBoost surrogates used to attribute the sCI-mediated SO_2 loss rate and its fractional share to controlling factors. The sCI loss rate is governed chiefly by **relative humidity and isoprene**; the drivers of the sCI fraction split diurnally — alkene speciation, O_3 and alkene share of VOCs by day, alkene speciation, RH and NO_x by night. Under high-sCI-fraction regimes, the positive response of total SO_2 oxidation to O_3 , VOCs and alkene fraction is *amplified*, which a summer episode at Wuhai supports. Policy consequence: where alkene ozonolysis is active, **cutting VOCs controls sulfuric acid formation more effectively than cutting SO_2 alone** — and sulfuric acid is the new-particle-formation precursor. Box model with a machine-learned surrogate; the surrogate inherits every mechanistic assumption underneath it.

doi: 10.5194/acp-26-12479-2026

Zhao H et al. 2026 Environ Monit Assess*Lead isotopes resolve dustfall sources across urban functional areas*

Ten trace elements (V, Mn, Co, Cr, Ni, Cu, Zn, As, Cd, Pb) in dustfall from **16 sites** across residential, busy-road, park and industrial areas of Baoding, with enrichment factors, geo-accumulation index and Pb isotope fingerprinting. Cu, Pb, Zn and Cd reach considerable pollution levels; the spatial ranking is **residential > busy road > industrial > park**, which is not the ordering most source-apportionment intuition predicts and the paper does not fully explain. Total non-carcinogenic risk stayed below threshold but **total carcinogenic risk exceeded the acceptable threshold** in several areas, worst on busy roads and in residential zones. Pb isotopes attribute **over 50%** to a combined coal-combustion plus traffic-emission group in most areas — combined, i.e. the isotope system does not separate the two here, which is the honest limit of the method in a city that burns coal and drives cars.

doi: 10.1007/s10661-026-15859-0

Lei et al. 2026 Atmos Environ

Source transition in PM_{2.5}-bound metals in a coastal megacity

Ships on metadata only — no abstract in the Crossref deposit and no other index carries the record on this run. The title indicates a multi-year, high-time-resolution observational record of PM_{2.5}-bound metals with a source-transition analysis and a health-implication component. Counted and listed; no numbers claimed. High retry priority: multi-year hourly metals is exactly the dataset class against which a low-cost network's composition proxies would have to be validated, and there are very few of them.

doi: [10.1016/j.atmosenv.2026.122336](https://doi.org/10.1016/j.atmosenv.2026.122336)

Mutic et al. 2026 medRxiv (preprint)

Paired silicone wristbands across home and early-childhood-education settings

A *protocol*, not results. Prospective repeated-measures pilot targeting **44 preschool children and 8 staff** across two sociodemographically distinct Atlanta early-childhood centres, with home- and centre-designated silicone wristbands exchanged between settings over three consecutive days and nights, one spot urine per child, and continuous indoor air-quality monitoring in two classrooms per site. Eight pre-specified feasibility indicators. The design premise is sound and underexploited — children spend >90% of time indoors and early-childhood centres are high-occupant-density microenvironments almost absent from the exposure literature — but wristbands integrate semi-volatile organics, not particle mass, so the PM component rests entirely on the fixed classroom monitors. Listed for the design; nothing to conclude from it yet.

doi: [10.64898/2026.08.27.26361470](https://doi.org/10.64898/2026.08.27.26361470)

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
De Potter et al.	Influences of socioeconomic and environmental factors on STEMI incidence and related mortality: insights from the Belgian Environmental Myocardial Infarction Risk (EMIR) study	<i>Eur Heart J Acute Cardiovasc Care</i>	Time-stratified case-crossover + DLNM	Acute and chronic residential PM _{2.5} , NO ₂ , O ₃ , temperature, noise and greenspace from high-resolution models; per IQR	Belgium
Reproductive & developmental					
Zanini et al.	Maternal exposure to air pollution and fetoplacental Doppler markers during pregnancy	<i>Environ Pollut</i>	Metadata only	Maternal residential air-pollution exposure during pregnancy	Europe
Respiratory & allergic					
Li H et al.	Multi-omics causal inference of childhood asthma triggered by ambient particulate matter	<i>ERJ Open Res</i>	Mendelian randomisation + TWAS	Genetically instrumented ambient PM _{2.5} from GWAS summary statistics	Europe
Occupational & indoor					
Chen Y et al.	Household solid fuel combustion-related PAH exposure in rural northern China: associations of phenanthrene metabolites with DNA damage and systemic inflammation	<i>Atmos Environ</i>	Metadata only	Household solid-fuel PAH exposure indexed by urinary phenanthrene metabolites	China
Lv H et al.	Structurally engineered polyimide microsphere/nanofibrous gradient aerogels with enhanced dust-holding capacity for high-temperature air filtration	<i>ACS Appl Mater Interfaces</i>	Laboratory instrument development	PM0.3 filtration efficiency, pressure drop and dust-holding capacity of a single-component polyimide gradient aerogel	Laboratory
Ma L et al.	Evaluating surgical smoke in laser-assisted laparoscopic partial nephrectomy with a novel handle: an in vitro study	<i>Can J Urol</i>	Bench / device experiment	Surgical-smoke formaldehyde, TVOC and CO with and without an integrated aspirator handle	Laboratory
Micallef & Scerri	A flow conditioned infiltration metric for traffic-related air pollution ingress and indoor air quality in naturally ventilated buildings	<i>Build Environ</i>	Metadata only	Outdoor-to-indoor infiltration of traffic-related air pollution, conditioned on flow regime	Europe
Sensing, forecasting & instrumentation					
Chaparro et al.	Monitoring airborne fine particles with biological and technological sensors for preventive conservation of cultural heritage sites	<i>Sci Total Environ</i>	Metadata only	Airborne fine particles measured in parallel by biomonitors and instrumental sensors	Latin America
Delkash	Receptor-scale imprint of barometric pressure variability on community air monitoring near an elevated-temperature landfill: a multi-method analysis across two mitigation interventions	<i>Environ Pollut</i>	Metadata only	Community-monitor receptor concentrations against barometric pressure variability, across two landfill mitigation interventions	United States
Exposure assessment & modelling					
Engeroff et al.	Air pollution exposure during urban running: a randomized crossover comparison of particulate matter exposure on park, roadside, and residential routes with consideration of rush-hour timing	<i>Research Square (preprint)</i>	Mobile monitoring campaign	Mobile personal PM ₁₀ , PM _{2.5} and PM ₁ during 2 km runs, referenced to local urban background PM _{2.5} /PM ₁₀	Germany

First author	Focus	Journal	Design	Metric	Region
Lei et al.	Source transition and health implication of PM _{2.5} -bound metals: multi-year high-resolution observation in a coastal megacity	<i>Atmos Environ</i>	Metadata only	Hourly PM _{2.5} -bound metals, multi-year record	China
Mutic et al.	Characterizing preschool children's multi-matrix air pollutant exposures across home and early childhood education settings: a paired silicone wristband study protocol	<i>medRxiv (preprint)</i>	Cross-sectional time-activity survey	Paired home- and centre-designated silicone wristbands plus continuous indoor air-quality monitoring in two classrooms per site	United States
Petersen et al.	Direct photolysis and photosensitized reactions of sulfate ions in aerosol water form radical oxidants, secondary brown carbon, and organosulfates	<i>Environ Sci Technol</i>	Bench / device experiment	Sulfate-radical (SO ₄ anion radical) production in concentrated aerosol water; transient absorption plus radical trapping with LC-HRMS	Laboratory
Zhao H et al.	Lead isotope characteristics and source analysis of dust fall in different functional areas of a typical city in northern China	<i>Environ Monit Assess</i>	Source-apportionment method	Dustfall trace elements (V, Mn, Co, Cr, Ni, Cu, Zn, As, Cd, Pb) with Pb isotope source fingerprinting	China
Zhu et al.	Revisiting the critical role of stabilized Criegee intermediates in sulfuric acid formation: coupling mechanistic updates with interpretable machine learning	<i>Atmos Chem Phys</i>	Chemical transport modelling	Gas-phase SO ₂ oxidation rate via stabilized Criegee intermediates; sCI fraction of total SO ₂ loss	China

Provenance and method

Entry window **2 September 2026**, single day, opened against a `last_entry_date` of 1 September. Sources queried on 3 September 2026; under this pipeline's 22:00-local rule, 2 September is the newest buildable date and 3 September is not opened. **PubMed** E-utilities on [EDAT] 2026/09/02: 4 hits on the health axis and 0 on the instrumentation axis. **Europe PMC** CREATION_DATE: 17 hits, including the two preprints carried here. **Crossref** by-ISSN, windowed on `created`: 68 deposits. **OpenAlex** HTTP 403 (plan-gated date filter) and **arXiv** HTTP 429; both legs recorded as failed, not empty. Consensus was not re-queried — it returned three already-seen records on the 1 September run and its calibration axis does not move daily. After collapsing cross-source duplicates by DOI and normalised title, 89 unique records were considered; **12 were already shipped** (11 on 1 September, 1 on 31 August — the by-ISSN sweep re-lists a deposit for several days), 15 were screened in scope and 62 were screened out. Notable exclusions: an *Atmos. Environ.* O₃-only cohort (10.1016/j.atmosenv.2026.122338) and a Barcelona indoor-NO₂ modelling study, both excluded as non-particulate; a settled-dust contaminants-of-emerging-concern survey, excluded as a chemical rather than aerosol measurement. **Six of the fifteen in-scope records ship on metadata only** and carry no attributed numbers; every one is on the abstract retry list. Subtopic, design, particle-metric and geography labels are single-label by dominant contribution and are a judgement, not a controlled vocabulary. All DOI links resolve to the publisher of record; abstracts remain the copyright of their respective publishers.